✎ **Teacher note (delete this box before publishing):** Replace every yellow-highlighted field with your own information (name, email, office hours, grading weights). Nothing else needs to change.

8th Grade Pre-Algebra

OPTIMA ACADEMY ONLINE · COURSE SYLLABUS · 2026-2027

Full Year · Grade 8

FACILITATOR

[Ms./Mr. Your Name]

EMAIL & OFFICE

[yourname@optimaed.com] · 239.673.2801 (OAO Front Office)

FLORIDA COURSE

[Florida 1205070 - confirm number] · Full-year course

OFFICE HOURS

[Day & time - e.g., Thursdays 3:00-4:00 PM] · weekly Study Hall

GRADE LEVEL

Grade 8 (Middle School)

FORMAT

On-Demand (also offered live on the VR campus)

About this syllabus. This is the plan for the course as designed. It is a full-year Grade 8 course; On-Demand scholars pace through the twelve modules across the four quarters of their enrolled year. On-Demand and live sections share the same content and expectations; exact pacing, scheduled dates, and a few activities may vary from section to section and term to term. Your facilitator will confirm the specifics for your section.

COURSE OVERVIEW

8th Grade Pre-Algebra builds the foundation for high school mathematics. Across twelve modules, scholars master the Laws of Exponents and scientific notation, extend the number system to include irrational numbers, and learn to generate and operate with algebraic expressions. From there the work turns to solving equations and inequalities, understanding functions and proportional relationships, and graphing and interpreting linear functions and systems. The final quarters move into data and probability and into geometry - angle relationships, the Pythagorean Theorem, and transformations with similar triangles. Every lesson pairs a short interactive lesson with independent practice, and each module includes checkpoints, a summative review, a summative assessment, and an immersive VR component that reinforces the concept. Throughout, scholars connect procedures to real-world reasoning, taking a crucial step toward Algebra, Geometry, and Statistics.

WHAT SCHOLARS WILL LEARN TO DO

- Apply the Laws of Exponents and use scientific notation to work with very large and very small quantities.
- Understand the real number system, including irrational numbers, and plot, order, and compare them.
- Generate equivalent algebraic expressions and operate with monomials and linear expressions.
- Solve multi-step linear equations and inequalities in one variable, and simple radical equations.
- Determine whether a relationship is a function and distinguish linear from nonlinear functions.
- Graph and interpret linear functions - slope, intercepts, and proportional relationships.
- Solve systems of two linear equations and reason about the number of solutions.
- Construct scatter plots and line graphs and describe patterns of association in bivariate data.
- Find theoretical probabilities and model repeated and simulated experiments.
- Reason about angle relationships, apply the Pythagorean Theorem, and analyze transformations and similar triangles.

MAJOR CHECKPOINTS & MILESTONES

The course builds toward a summative assessment at the end of each module. The quarter checkpoints below follow a sample full-year pacing; On-Demand scholars pace to their own enrolled term and live sections follow the class schedule, keeping each checkpoint roughly in order.

QUARTER / MODULES	FOCUS	SAMPLE TARGET	CHECKPOINT
Q1 · Modules 1-3	Exponents & scientific notation, real numbers, algebraic expressions	by Oct 9, 2026	Module 1-3 summative assessments
Q2 · Modules 4-6	Equations & inequalities, functions, linear functions	by Dec 18, 2026 *(Semester 1 ends)*	Module 4-6 summative assessments
Q3 · Modules 7-9	Systems of equations, bivariate data, probability	by Mar 5, 2027	Module 7-9 summative assessments
Q4 · Modules 10-12	Angle relationships, triangles & Pythagorean Theorem, transformations	by May 28, 2027 *(course ends)*	Module 10-12 summative assessments

IMPORTANT DATES - 2026-2027

Aug 12, 2026	First day of school	**Sep 7**	Holiday (Labor Day)
Oct 9	Quarter 1 ends	**Oct 12**	Holiday (Columbus Day)
Nov 11	Holiday (Veterans' Day)	**Nov 23-27**	Thanksgiving break
Dec 18	Quarter 2 ends · fall semester ends	**Dec 21-Jan 1**	Winter break

Jan 5, 2027	Classes resume	**Jan 18**	Holiday (MLK Day)
Feb 15	Holiday (Presidents' Day)	**Mar 5**	Quarter 3 ends
Mar 26-Apr 5	Spring break	**Apr 6**	Classes resume
May 28, 2027	Quarter 4 ends · last day of school	**May 31**	Holiday (Memorial Day)

School calendar dates follow Optima's board-approved 2026-2027 calendar and may be adjusted by the school.

ASSESSMENT & GRADING

Suggested default weighting - set these to match your section and your school's grading policy. Every practice item allows unlimited attempts so scholars can review and try again; summative assessments allow two attempts.

40%
Independent Practice & Mid-Module Checks

40%
Module Summative Assessments

20%
Summative Reviews & VR Activities (completion)

HOW THIS COURSE IS DELIVERED

This course is offered in two formats that share the same content, assignments, and expectations: an **On-Demand (asynchronous)** section you pace yourself, and a **Live (synchronous)** section that meets on a schedule on Optima's VR campus. Find the one that matches your enrollment.

On-Demand (Asynchronous) Learning

In the On-Demand section you move through the course at your own pace within each quarter, guided by due dates that keep you on track. All assignments for a quarter open at its start and close at its end. There is no grade penalty for missing an individual due date - but any assignment not completed by the end of the quarter receives a zero, and assignments submitted in the final 14 days of a quarter receive a 10% deduction, so steady pacing matters.

Attendance is measured by assignment completion. Per school policy, a scholar who completes no assignment for ten consecutive school days, without notifying the school in advance, may be dropped from the course. Completion means a sincere attempt - real effort on real work. Rushed work, wrong or corrupted files, nonsense submissions, and AI-generated filler do not count.

You are never on your own. A live facilitator grades your work, gives feedback, and hosts a weekly Study Hall.

Live (Synchronous) Learning

In the Live section you join scheduled class meetings on Optima's VR campus, where your teacher leads instruction in real time and you learn alongside classmates. The lessons, assignments, and assessments are the same as the On-Demand course; what differs is the pacing - a Live section moves through the material together on the class schedule, and class time is also used for discussion, questions, and guided practice.

Attendance is measured by presence and participation. You are expected to attend scheduled sessions on time and ready to take part. If you must miss a class, notify the school in advance; your teacher will let you know how to make up the work.

🎮 **You meet regularly with a teacher.** The class gathers throughout the week in VR and on Microsoft Teams for live instruction and discussion.

APPROPRIATE USE OF AI

Optima scholars are preparing for a world shaped by artificial intelligence, and learning to use these tools with good judgment is part of that preparation. You are welcome to use AI tools to support your learning - to explain a concept a different way, check your understanding, or brainstorm - unless a particular assignment says otherwise. What AI may never do is the thinking or the work an assignment asks of you: your reasoning on a problem, a reflection, or a written explanation must be genuinely your own. In math especially, the goal is for you to be able to judge whether an answer is right, which you can only do if you can do the work yourself. When you do use an AI tool on an assignment, say so briefly.

LEARNING IN VIRTUAL REALITY

Optima is a virtual-reality school. Alongside your Canvas coursework, you will join immersive sessions on Optima's VR campus, where you meet teachers and fellow scholars, take part in live activities, and explore mathematical ideas in ways you could not in an ordinary classroom. Each lesson includes a VR component that reinforces the concept. Keep your headset charged and ready for any scheduled VR sessions, and treat the VR campus with the same courtesy you would a physical one. If you ever cannot get into VR for a session, tell your facilitator - there is an alternative way to take part so you do not fall behind.

BEING AN OPTIMA SCHOLAR

Optima's mission is an excellent, virtue-centered education, and virtue is at the center of this course as much as any other. Every scholar is expected to practice the seven virtues that shape life at Optima - **courage, courtesy, honesty, perseverance, self-government, service, and responsibility**. In practice that means: do your own honest work; treat everyone you meet in, through, or because of this course with respect; engage actively rather than passively; come to Study Hall to get help and get to know your facilitator; and take responsibility for your own pacing and progress. You belong to a house and a community of scholars, not only to a course.

WHAT YOU NEED & STAYING CONNECTED

- A computer with reliable internet and access to **Canvas**.
- A **VR headset**, charged and ready for the immersive activities on Optima's VR campus.
- The **Grade 8 Reference Sheet** and a state-approved online calculator (both linked in the Start Here section).
- **Microsoft 365** (Word, Excel) with OneDrive, for any documents you create and submit.
- A notebook and pencils for working problems by hand - the most important math tool you own.

Staying connected. Updates about this course are posted on the course's Canvas page; school-wide updates are posted in your Homeroom Canvas course. Check both regularly. Your facilitator holds office hours and is glad to help with the material - come by, or reach out any time by email.

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